Exact planar Chambolle–Pock three-cycles and an optimal ordered triangle family

Abstract

For every nonnegative extrapolation parameter, we construct an exact period-three orbit of the literal primal-first Chambolle–Pock map using two triangles in ordinary Euclidean space. We determine the smallest stepsize product admitting the prescribed orbit within an explicitly restricted ordered equilateral-triangle class. The proof reduces the projection conditions to four scalar inequalities and an exact ellipse support calculation. At extrapolation $1/8$, the product is $(88+16\sqrt{91})/81 \approx 2.970744$, below the quadratic benchmark $16/5$. More generally, the family lies below that benchmark on an exact interval of extrapolation parameters. The result does not determine the universal convex convergence threshold. No novelty claim is made.

Keywords: Chambolle–Pock iteration; period-three orbits; equilateral triangles; convex projection; restricted product threshold

2020 Mathematics Subject Classification. Primary 39A23; Secondary 37N40, 90C25.

1 Recurrence and exact planar data

All inner products and norms below are the ordinary Euclidean ones on $\mathbb{R}^2$. The primal-first recurrence is

$$\begin{aligned} x_{n+1} &= \operatorname{prox}_{\tau f}(x_n - \tau L^* y_n), \\ y_{n+1} &= \operatorname{prox}_{\sigma h}\big(y_n + \sigma L((1+t)x_{n+1} - t x_n)\big), \end{aligned} \tag{1}$$

where $h = g^*$. There is no outer relaxation or extra memory. We use the primal–dual framework of Chambolle and Pock [1], while specifying the update order and initialization explicitly.

Fix $t \geq 0$ and put

$$W = 1 + 3t + 3t^2, \qquad s = \sqrt{W}, \qquad \Delta = 2s - 1 - 3t, \qquad r_t = \frac{3}{\Delta}. \tag{2}$$

Let R be counterclockwise rotation by 120 degrees:

$$R = \begin{pmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{pmatrix}.$$

Define, with indices read modulo three,

$$c_i = R^i(1,0), \qquad d_i = R^i\left(\frac{1+t}{2}, -\frac{\Delta}{2\sqrt{3}}\right), \qquad i = 0,1,2. \tag{3}$$

Set $C = \operatorname{conv}\{c_0, c_1, c_2\}$, $D = \operatorname{conv}\{d_0, d_1, d_2\}$, $f = \iota_C$, $h = \iota_D$, $L = I$, $\sigma = 1$ and $\tau = r_t$. Here ι_E is zero on E and $+\infty$ elsewhere, and $g(z) = \max_{0 \leq i \leq 2} \langle z, d_i \rangle$ is the support function of D.

2 The cycle and its precise optimality scope

Theorem 1. *For every $t \geq 0$, $\Delta > 0$, and the data above are proper closed convex with a saddle point at $(0,0)$ and $\|L\| = 1$. Starting at $(x_0, y_0) = (c_2, d_2)$, recurrence (1) has the exact orbit*

$$(x_{3k}, y_{3k}) = (c_2, d_2), \quad (x_{3k+1}, y_{3k+1}) = (c_0, d_0), \quad (x_{3k+2}, y_{3k+2}) = (c_1, d_1). \tag{4}$$

Its least period is three, and it does not converge. Its stepsize product is

$$\tau\sigma\|L\|^2 = r_t = \frac{2\sqrt{1+3t+3t^2}+1+3t}{(1+t)^2}. \tag{5}$$

Moreover, r_t is the smallest positive product permitting this ordered orbit in the following class: $c_i = R^i(1,0)$, $d_i = R^i(u,v)$ for arbitrary real u, v, indicator functions of their convex hulls, $L = I$, $\sigma = 1$, $\tau = r > 0$, and the fixed extrapolation t. The demanded transitions are $(c_{i-1}, d_{i-1}) \mapsto (c_i, d_i)$ for every i.

2.1 Positivity, saddle point, and projection criterion

The exact identity

$$4W - (1+3t)^2 = 3(1+t)^2 > 0$$

proves $2s > 1+3t$, hence $\Delta > 0$. Rationalizing $3/\Delta$ gives (5). Both triangles are nonempty compact convex sets. Their vertex sums vanish because $I + R + R^2 = 0$, so $0 \in C \cap D$. Thus

$$-L^*0 = 0 \in N_C(0), \qquad L0 = 0 \in N_D(0).$$

These are the KKT conditions, proving the saddle assertion. Equivalently, $f(x) + \langle x, y\rangle - h(y)$ has a saddle at $(0,0)$. The identity coupling has Euclidean operator norm one.

For a closed convex set E, the projection criterion is

$$v = P_E(z) \quad \Longleftrightarrow \quad v \in E, \quad \langle z - v, w - v\rangle \leq 0 \quad (w \in E). \tag{6}$$

Indeed, expand $\|z-w\|^2$ about v for sufficiency; for necessity, compare v with $v+\varepsilon(w-v)$ and let $\varepsilon \downarrow 0$. Strict convexity of squared distance gives uniqueness. On a triangle, checking the vertices suffices by affinity. Since $\operatorname{prox}_{a\iota_E} = P_E$ for every $a > 0$, this criterion applies directly to (1).

3 All phase inequalities and exact minimization

For an arbitrary member of the stated class, write

$$a = \langle c_0, d_0 \rangle = u, \qquad b = \langle c_0, d_1 \rangle = -\frac{u + \sqrt{3}v}{2},$$

$$\beta = \|d_0\|^2 = \frac{4}{3} q(a, b), \quad q(a, b) = a^2 + ab + b^2.$$

Every real pair (a, b) corresponds to exactly one (u, v). Rotational symmetry gives diagonal cross-pairings a, next cross-pairings b, and the remaining cross-pairings $-a - b$. Off-diagonal within-triangle pairings are $-1/2$ and $-\beta/2$.

For the transition to phase i, the actual proximal normals are

$$A_i = \frac{c_{i-1} - c_i}{r} - d_{i-1}, \qquad B_i = d_{i-1} - d_i + (1 + t)c_i - tc_{i-1}.$$

The primal residual is rA_i; division by positive r preserves its signs. Direct inner products with target differences in cyclic order $i + 1, i + 2$ give

	$i+1$	$i+2$
$\langle A_i, c_j - c_i \rangle$	$\frac{3}{2r} - a - 2b$	$\frac{3}{r} - 2a - b$
$\langle B_i, d_j - d_i \rangle$	$\frac{3\beta - 2a + (6t+2)b}{2}$	$3\beta - (3t+2)a - b$

(7)

The diagonal tests vanish. These four quantities being nonpositive is necessary and sufficient for all three ordered transitions, by (6).

3.1 The ellipse support bound

The first dual test and $3\beta = 4q(a, b)$ require

$$4q(a, b) \le 2a - (6t + 2)b.$$

Put $a_* = (1 + t)/2$ and $b_* = -(1 + 2t)/2$. Completing the square gives exactly

$$q(a - a_*, b - b_*) \le W/4. \tag{8}$$

For all real z, w, $4q(z, w) - (z + 2w)^2 = 3z^2 \ge 0$. Consequently (8) implies

$$a + 2b \le a_* + 2b_* + \sqrt{W} = \sqrt{W} - \frac{1 + 3t}{2} = \frac{\Delta}{2}.$$

The first primal test requires $a + 2b \ge 3/(2r)$. Thus every ordered cycle in the entire stated class must satisfy $r \ge 3/\Delta = r_t$. No sign assumption on a or b has been used.

3.2 Attainment and the initialized orbit

Choose

$$a = a_* = \frac{1+t}{2}, \qquad b = b_* + \frac{s}{2} = \frac{s-1-2t}{2}.$$

Then $u = a$ and $v = -(a+2b)/\sqrt{3} = -\Delta/(2\sqrt{3})$, exactly as in (3). This point attains equality in the ellipse support bound. At $r = r_t$, the four tests (7) become

$$0, \qquad 3b, \qquad 0, \qquad \frac{3}{2}\big[W - (1+2t)s\big]. \tag{9}$$

For the last equality, eliminate 3β using the first dual equality to obtain $-3[ta+(2t+1)b]$. Substitution then gives the displayed expression. Since

$$(1+2t)^2 - W = t(1+t) \geq 0,$$

we have $s \leq 1+2t$, so $b \leq 0$ and both remaining tests are nonpositive. This proves attainment and all three phase projections, including at $t = 0$, where all four tests are equalities. Uniqueness of each projection still follows from squared-distance minimization.

Starting at (c_2, d_2), the first primal projection produces c_0; the subsequent dual projection uses this newly produced c_0 and gives d_0. Repeating the verified transitions proves (4) by induction. The three primal vertices are distinct, so the least period is three. Distinct constant phase subsequences exclude convergence, also weak convergence in finite dimension. This completes the proof of Theorem 1.

4 Exact comparison with the quadratic benchmark

Proposition 4.1. The family satisfies

$$r_t < \frac{4}{1+2t} \quad \Longleftrightarrow \quad 0 \leq t < \frac{4\sqrt{5}-5}{22}. \tag{10}$$

Proof. All denominators are positive. The comparison is equivalent successively to

$$3(1+2t) < 4\Delta, \qquad 7 + 18t < 8\sqrt{W}.$$

Both sides of the latter comparison are positive, so squaring preserves equivalence. Using

$$(7+18t)^2 - 64W = 3(44t^2 + 20t - 5)$$

gives $44t^2 + 20t - 5 < 0$. Its positive root is $(4\sqrt{5}-5)/22$ and its other root is negative, proving (10) for $t \geq 0$. ∎

5 The special case $t = 1/8$

At $t = 1/8$, the exact parameters reduce to

$$r_{1/8} = \frac{88 + 16\sqrt{91}}{81} \approx 2.970744, \quad d_0 = \left(\frac{9}{16}, \frac{11 - 2\sqrt{91}}{16\sqrt{3}}\right), \quad c_0 = (1, 0).$$

The nonzero entries of (9) are

$$\frac{3(\sqrt{91} - 10)}{16} < 0, \qquad \frac{273 - 30\sqrt{91}}{128} < 0.$$

The signs follow from $\sqrt{91} < 10$ and $\sqrt{91} > 91/10$. Moreover

$$r_{1/8} < 3 < \frac{16}{5} = \frac{4}{1 + 2(1/8)}.$$

The first strict inequality is equivalent to $\sqrt{91} < 155/16$, verified by $91 \cdot 256 = 23296 < 24025 = 155^2$. Thus the planar family gives a convex projection obstruction below the quadratic benchmark, with an ordinary initialization and a saddle point.

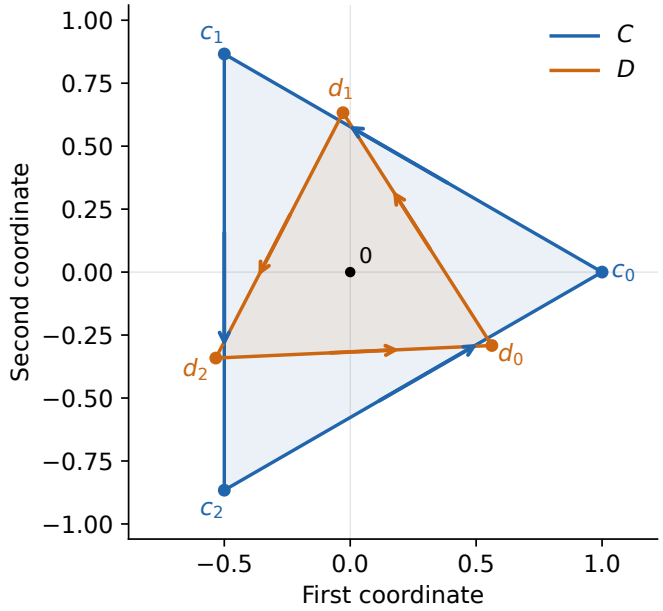


Fig. 1. The primal triangle C (blue) and dual triangle D (orange) at $t = 1/8$ in their common Euclidean plane. Arrows indicate the phase order $2 \to 0 \to 1 \to 2$ for each component; paired labels denote the joint states. The arrows are schematic phase transitions, not the unprojected update inputs. The plot illustrates the exact construction and is not used as a proof.

6 Scope and reproducibility

The optimality assertion in Theorem 1 concerns only the fixed ordered equilateral-triangle class stated there. It does not range over all convex functions, all dimensions, all couplings, all stepsizes with a prescribed product, or other types and orderings of trajectories. In particular, it does not identify the sharp universal convex convergence threshold.

The expression $4/(1+2t)$ is used here as a quadratic or bilinear benchmark. Formula (10) rules out its proposed unrestricted extension as a full-iterate convergence guarantee for all proper closed convex instances with a saddle point, at every parameter in that interval. No such extension is attributed to a published theorem. No conclusion about an algorithm with different update order, extrapolation, relaxation, or initialization is implicit.

These cycles are bounded, and their ordinary averages converge to the saddle point. Indeed, each consecutive block of three joint states sums to $(0, 0)$, and the finite remainder divided by the number of iterates tends to zero. Thus the construction is not an obstruction to boundedness or ordinary Cesàro convergence. Nor does the proof assert attraction or robustness under perturbed initial conditions.

6.1 Exact symbolic verification

The accompanying `verify.py` uses Python 3 and SymPy. Run `python3 verify.py`. It constructs the rotation and arbitrary dual vertex symbolically, derives the complete phase-normal tests directly from the update inputs, checks the ellipse completion and its support identity, and verifies the proposed parameters and all four attained expressions. It also checks the positivity and comparison polynomial identities and the exact $t = 1/8$ specialization.

The inequalities for arbitrary $t \geq 0$ follow from the explicit square identities in the proof; they are not inferred from numerical sampling or an optimization solver. Floating-point values appear only as explanatory approximations and in the illustrative plot. The figure can be regenerated by `python3 plot.py`. Run `bash build.sh` to compile the manuscript with a standard pdfLaTeX installation. The source package contains the manuscript, its style file, the symbolic verifier, the plotting script, and the vector figure.

References

[1] A. Chambolle and T. Pock, A first-order primal-dual algorithm for convex problems with applications to imaging, *Journal of Mathematical Imaging and Vision* **40** (2011), 120–145. doi:10.1007/s10851-010-0251-1.

A Exact range for the fixed metric construction

A related fixed construction has an exact parameter range distinct from the planar optimization above. In $\mathbb{R}^4$ use $\langle z, w\rangle_Q = z^{\mathsf{T}} Q w$, where

$$Q = \begin{pmatrix} 476 & -238 & 266 & -14 \\ -238 & 476 & -252 & 266 \\ 266 & -252 & 190 & -95 \\ -14 & 266 & -95 & 190 \end{pmatrix}.$$

Its leading principal minors are 476, 169932, 286552, and 362404, so Q is positive definite. Set

$$p_0 = e_1, \quad p_1 = e_2, \quad p_2 = -e_1 - e_2, \qquad q_0 = e_3, \quad q_1 = e_4, \quad q_2 = -e_3 - e_4.$$

Let $E = \operatorname{conv}\{p_i\}$ and $F = \operatorname{conv}\{q_i\}$. Use $f = \iota_E$, $h = \iota_F$, identity coupling in this common Hilbert norm, $\sigma = 1$, and $\tau = r > 0$. Allow the extrapolation t to be any real number.

The ordered transitions $(p_{i-1}, q_{i-1}) \mapsto (p_i, q_i)$ occur if and only if

$$r \geq 3, \qquad t \geq 5/42. \tag{11}$$

To check both directions, define the actual proximal normals

$$U_i = (p_{i-1} - p_i)/r - q_{i-1}, \qquad V_i = q_{i-1} - q_i + (1+t)p_i - t p_{i-1}.$$

Multiplication by Q gives, for every i, the two off-diagonal entries in cyclic target order:

$$\begin{aligned} \langle U_i, p_{i+1} - p_i\rangle_Q &= -238(r-3)/r, \\ \langle U_i, p_{i+2} - p_i\rangle_Q &= -14(37r - 102)/r, \\ \langle V_i, q_{i+1} - q_i\rangle_Q &= 5 - 42t, \\ \langle V_i, q_{i+2} - q_i\rangle_Q &= 52 - 798t. \end{aligned}$$

The diagonal entries vanish. The projection criterion requires exactly that these four quantities be nonpositive. Since $3 > 102/37$ and $5/42 > 26/399$, the conditions reduce to (11).

Both vertex sums are zero, so $(0, 0)$ is a saddle point. Initialization at (p_2, q_2) yields the exact period-three orbit whenever (11) holds. The identity norm is one. In particular, $r = 3$ lies strictly below $4/(1 + 2t)$ exactly for $5/42 \leq t < 1/6$ within this admissible range. Equality holds at $t = 1/6$. This is a statement about this fixed construction, not a universal threshold. The accompanying verifier checks Q and every phase table exactly.